

Counting Methods

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Distinct and identical objects

Objects are **distinct** if we can identify each one separately. Objects of the same type are **identical** for a counting problem if exchanging them leaves the outcome unchanged.

The letters A, B, C, D have $4! = 24$ arrangements. But A, B, C, C have fewer arrangements because swapping the two copies of C leaves the same word.

We first count arrangements with fixed numbers of repeated symbols, then distributions of identical objects.

Arranging $ABCC$

How many different strings use all four letters in $ABCC$ exactly once? Temporarily label the copies of C as C_1, C_2 . The four distinct labeled letters have $4!$ permutations.

$$ABC_1C_2 \quad \text{and} \quad ABC_2C_1$$

both become $ABCC$ after we erase the labels.

Every visible arrangement has exactly $2!$ labeled versions, so

$$\frac{4!}{2!} = 12$$

different strings remain.

Multinomial coefficients

Suppose n objects have r types, with n_i identical copies of type i .

Here $n_1, \dots, n_r$ are nonnegative integers and $n_1 + \dots + n_r = n$.

Labeling every copy gives $n!$ permutations. Each visible arrangement has $n_1! \cdots n_r!$ labeled versions.

Therefore the number of arrangements is the **multinomial coefficient**

$$\binom{n}{n_1, \dots, n_r} = \frac{n!}{n_1! \cdots n_r!}.$$

For two types, this becomes $\binom{n}{n_1}$, since $n_2 = n - n_1$.

Arranging $AABBC$

How many strings contain exactly two A 's, two B 's, and one C ?

The multiplicities are 2, 2, 1, giving

$$\frac{5!}{2!2!1!} = 30.$$

We can also choose the positions: two of five for A , then two of the remaining three for B . The last position contains C .

$$\binom{5}{2} \binom{3}{2} = 10 \cdot 3 = 30.$$

This also explains the count without labeling identical copies.

Groups of prescribed sizes

Divide 9 distinct students into labeled groups A, B, C of sizes 2, 3, 4.
Order within a group does not matter.

Choose A , then B , then put the remaining students in C :

$$\binom{9}{2} \binom{7}{3} \binom{4}{4} = \frac{9!}{2!3!4!} = 1260.$$

Multinomial coefficients also count allocations of distinct objects to labeled groups of fixed sizes. The group labels matter, so we do not divide by the number of group orders.

Distributing identical objects

Distribute 8 identical tokens among three labeled boxes A, B, C . Empty boxes are allowed. How many distributions are possible?

Let x_1, x_2, x_3 be the numbers of tokens in these boxes. We count integer solutions of

$$x_1 + x_2 + x_3 = 8, \quad x_1, x_2, x_3 \geq 0.$$

The outcomes $(3, 0, 5)$ and $(0, 3, 5)$ differ because the boxes have labels. Within a box, exchanging tokens makes no difference.

Encoding a distribution

Represent each token by a star and separate the three boxes with two bars. The distribution $(3, 0, 5)$ corresponds to

$$\underbrace{***}_A \mid \underbrace{}_B \mid \underbrace{*****}_C.$$

Adjacent bars mean an empty box. A bar at the beginning or end means the first or last box is empty.

Each distribution determines one string of eight stars and two bars, and each such string determines one distribution.

Stars and bars

With $n \geq 0$ identical objects and $k \geq 1$ labeled boxes, use n stars and $k - 1$ bars.

There are $n + k - 1$ positions. Choose the $k - 1$ positions occupied by bars:

$$\boxed{\binom{n + k - 1}{k - 1}}.$$

This counts the integer solutions of $x_1 + \cdots + x_k = n$ with every $x_i \geq 0$.

For eight tokens and three boxes, the number is

$$\binom{8 + 3 - 1}{3 - 1} = \binom{10}{2} = 45.$$

At least one object in every box

Now require every one of the k labeled boxes to receive at least one of the n identical objects, where $n \geq k \geq 1$.

Give each box one object first. Let $y_i = x_i - 1$ count its remaining objects. Then

$$y_1 + \cdots + y_k = n - k, \quad y_i \geq 0.$$

Stars and bars gives

$$\binom{(n - k) + k - 1}{k - 1} = \boxed{\binom{n - 1}{k - 1}}.$$

If $n < k$, no distribution can give every box an object.

Nine identical gifts

Give 9 identical gifts to three named children. How many distributions give each child at least one gift?

After giving one gift to each child, distribute the remaining 6 freely:

$$\binom{6+3-1}{3-1} = \binom{8}{2} = 28.$$

If a child may receive no gifts, the count instead is

$$\binom{9+3-1}{3-1} = \binom{11}{2} = 55.$$

The minimum allowed amount changes the counting problem.

Which objects are distinguishable?

Distribute 8 objects among three labeled boxes, allowing empty boxes.

Distinct objects: each object independently chooses a box.

$$3^8 = 6561.$$

Identical objects: only the three numbers of objects matter.

$$\binom{10}{2} = 45.$$

Swapping distinct objects between boxes can create a different outcome. Swapping identical ones cannot.

Set operations

For subsets A, B of a finite set U :

- ▶ $A \cup B$ is the **union**: elements in at least one of A, B .
- ▶ $A \cap B$ is the **intersection**: elements in both A and B .
- ▶ $U \setminus A$ is the **complement of A in U** : elements of U outside A .

Recall that $|A|$ denotes the number of elements of A . Counting A and everything outside it gives

$$|U \setminus A| = |U| - |A|.$$

Two-set inclusion–exclusion

For finite sets A, B , the sum $|A| + |B|$ counts each element of $A \cap B$ twice. Subtract that extra copy:

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}.$$

For example, 20 students study Python, 15 study Java, and 8 study both. The number studying at least one language is

$$20 + 15 - 8 = 27.$$

When the sets are disjoint, the intersection is empty and we obtain the sum rule.

Three-set inclusion–exclusion

For finite sets A, B, C ,

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|. \end{aligned}$$

Add the three set sizes and subtract the pairwise intersections.

An element in all three sets has then contributed $3 - 3 = 0$. Adding the triple intersection counts it once, as required.

Counting each element once

Track one element through the three-set formula.

Number of sets containing it	Total contribution
0	0
1	1
2	$2 - 1 = 1$
3	$3 - 3 + 1 = 1$

Every element in the union contributes exactly one. Every element outside it contributes zero. This proves the formula.

Strings containing every symbol

How many strings of length 5 over $\{A, B, C\}$ contain **each of** A, B, C **at least once**?

Repetition is allowed: $AABBC$ qualifies, but $ABABA$ does not because it has no C .

Let U be the set of all length-5 strings over $\{A, B, C\}$, without the requirement to use every symbol. There are

$$|U| = 3^5 = 243.$$

We will subtract the strings that miss at least one of the three symbols.

Missing-symbol sets

Define subsets of U by

$$M_A = \{\text{strings with no } A\}, \quad M_B = \{\text{strings with no } B\},$$

and $M_C = \{\text{strings with no } C\}$.

Each set has 2^5 elements because each position has two symbols available.

Each pairwise intersection has one element. For example,

$$M_A \cap M_B = \{CCCCC\}.$$

The triple intersection is empty: a length-5 string cannot miss all three available symbols.

Applying inclusion–exclusion

The strings missing at least one symbol form $M_A \cup M_B \cup M_C$.
Its size is

$$|M_A \cup M_B \cup M_C| = 3 \cdot 2^5 - 3 \cdot 1 + 0 = 93.$$

Therefore the number containing every symbol is

$$|U| - |M_A \cup M_B \cup M_C| = 243 - 93 = \boxed{150}.$$

Counting the missing-symbol cases avoids splitting into all possible positions and multiplicities of A, B, C .

Pigeonhole principle

Let $m \geq 1$ be an integer.

Pigeonhole principle

If $m + 1$ objects occupy m boxes, some box contains at least two objects.

A box can mean any category, such as a color or a month.

Among 13 people, at least two share a birth month: there are only 12 possible months. This conclusion requires no assumptions about how likely the months are.

Why the pigeonhole principle holds

Suppose every one of the m boxes contained at most one object.
Then the total number of objects could be at most

$$\underbrace{1 + \cdots + 1}_{m \text{ boxes}} = m.$$

This contradicts the presence of $m + 1$ objects. Therefore some box contains at least two.

The argument guarantees repetition without determining which objects repeat.

Generalized pigeonhole principle

Write $\lceil x \rceil$ for the least integer greater than or equal to x . For integers $N \geq 1$ and $m \geq 1$, distributing N objects among m boxes puts at least

$$\left\lceil \frac{N}{m} \right\rceil$$

objects in some box.

Otherwise each box contains at most $\lceil N/m \rceil - 1 < N/m$ objects, so the total is less than N , a contradiction.

Among 51 people, some birth month contains at least $\lceil 51/12 \rceil = 5$ people.

Summary

- ▶ **Repeated types:** divide labeled arrangements by the number of labels producing each outcome.
- ▶ **Identical objects in labeled boxes:** use stars and bars, checking whether empty boxes are allowed.
- ▶ **Overlapping cases:** use inclusion–exclusion, often together with counting the complement.
- ▶ **Guaranteed repetition:** use the pigeonhole principle.